

Tunisia, audited — the engineer’s ledger

the energy balance, water chemistry, acid-stream assays, lift floors, and every gate
— everything a technical team verifies before FID.

Provenance law: MEASURED = cited anchor · DERIVED = arithmetic on measured values · ASSUMED = declared default. commercially-validated = 0 — no buyer has signed anything; published as zero on purpose. Generated 2026-09-09 by nations_audit.py / make_nations_pdfs.py — the same code that renders the page.

1. The energy ledger, leg by leg

Recovery ledger recomputed: 33.9 GWh/yr recovered + 20 GWh/yr dispatch value. The printed 82 GWh did not sum from its own legs (7.2 + 30 + 20 = 57.2; 24.8 unexplained) — corrected, with the arithmetic, on the page.

LEG	FORMULA	VALUE
Pump-as-turbines	$9.81 \times (70\% \times 150,000/86,400) \text{ m}^3/\text{s} \times 50 \text{ m} \times 75\% \times 8,760 \text{ h}$	3.9 GWh/yr
Heat recovery	$0.6 \text{ GJ}/\text{t} \times 2 \text{ Mt PG} \rightarrow 333 \text{ GWh thermal} \times 9\%$	30 GWh/yr
Flexibility	desal + storage follow the solar curve	20 GWh/yr (dispatch, not joules)
ERD	inside the 4 kWh/m ³ SWRO benchmark (~35-40% of RO energy)	0 added — no double count

2. Water — the numbers an EPC checks first

PARAMETER	VALUE	ANCHOR
SWRO specific energy	4 kWh/m ³ (benchmark incl. ERD)	Rw3 (mega-plant class)
Lift physics floor	1.09 kWh/m ³ per 400 m at 100% eff; 1.45 at 75%	Rw4 — the physics that decides where water may travel
Delivered inland cost	\$1.10-1.50/m ³ vs irrigation tariffs \$0.04-0.07	Rw6/Rw7 — inland water runs on brackish, not long-haul
SWRO CAPEX	\$800-1,200 per m ³ /day (mega-plant)	Rw2
Gate	blended ≥\$0.68/m ³ , ≥60% pre-sold, brine permit	computed (DSCR)

3. Materials — the acid stream

PARAMETER	VALUE	ANCHOR
REE content, Gafsa rock	322 mg/kg avg, peaks >1,700	Rm1
PG partition of REE	60-85%	Rm2
Acid-stream U	85-90% of rock-U partitions to the wet-process acid	Ru1 (Freeport precedent)
U recovery (proven)	>95% DEHPA/TOPO, 14M lbs over 1978-99	Ru1
Radiology	Ra-226 214-970 Bq/kg in PG; exemption threshold 1,000 Bq/kg	Rm11/Rm12
Phase 0 assays	12 stream points: U, REE, Ra-226, Po-210, full metals	itemized in the audit

4. The dependencies, by status

DEPENDENCY	STATUS
Desal brine permit	in design — Gabes outfall / diffuser; permit required before SWRO FID
PG chemistry (Ra-226 partitioning)	Phase 0 assay — flowsheet keeps radium in gypsum (Freeport precedent); re-verify on Gafsa stream
Uranium handling	gate — CNSTN licensing, NORM transport, export controls
REE separation capex	later phase — +\$244M solvent-extraction plant beyond the ~\$100M circuit
Solar bankability	gate — DSCR ~0.9-1.0 at the record-low tariff; bankable region or grant support required
Grid connection	later phase — 3 GW connection + 400 kV upgrades ~\$450M
Compute accelerators	gate — export-control licenses; colocation until an anchor tenant is licensed
Truffle yield	gate — 376 kg/ha is the 20-yr Murcia average; gate = 3-season record ≥200 kg/ha
Export prices	exposure — U spot \$80-95/lb; truffle EUR20-40/kg — no contracted offtake yet
Financing	uncommitted — DFI channel modeled for solar/desal; nothing committed

The water stack — line-by-line, with each credit's failure mode

The \$0.347/m³ audit: the base is measured anchors; each credit is a loop-internal yield with a named failure mode; the degraded cascade still clears the \$0.68 gate.

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The people's energy is electrons, not hydrogen — and the H&sb2; lives inside the loop

The loop's H&sb2; (2.63 kt/yr) is an industrial molecule: per kWh of heat, green H&sb2; at \$0.21 is 7× the loop's \$0.031/kWh electron; blending caps at 5-20% by volume (Pg20/Pg21/Pg22); the whole H&sb2; volume equals 0.0% of grid-gas demand. The molecule goes where only a molecule works: ammonia for the GCT chain, inside the loop. the electrolyzer sits on the desal line, and everything it touches stays in the loop: the water it splits is the SWRO permeate (0.04% of the 150,000 m³/d output); its un-split reject stream is clean water that returns to the product line; its oxygen by-product aerates the spirulina ponds and greenhouses; its stack heat (15–20% of input) joins the kiln and DC reject heat on the desal preheat circuit; and the minerals concentrated in its flush stream join the brine valorization line. The electrolyzer is not a consumer of the loop — it is another recovery point inside it.

What this document is: a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

What it is not: a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

Works cited

- [Rm1] Rare Earth Elements in Phosphate Ores and Industrial By-Products (Gafsa avg 322 mg/kg, peaks >1,700) — <https://www.mdpi.com/2075-163X/15/12/1232...>
- [Rm2] Global phosphogypsum: generation, stockpiles and 60-85% REE partition to PG — <https://www.mdpi.com/2071-1050/17/19/8828...>
- [Rm3] Gabès phosphate plants REE characterization: enrichments to 1,075 mg/kg; Ce 80-309 mg/kg; negligible Dy/Tb — https://www.researchgate.net/publication/352414505_Rare_earth_elements_characterization_associated_to_the_phos...
- [Rm4] Characterization and Leaching Kinetics of REE from Phosphogypsum in HCl (1.65 mol/L; 65.6% ΣREE; 73.8% Y) — https://www.researchgate.net/publication/361001660_Characterization_and_Leaching_Kinetics_of_Rare_Earth_Elemen...

5. [Rm5] Recovering REE from phosphogypsum as a secondary resource: scale-up failure modes (filtration, silica gel, co-extraction) — <https://www.chinanhd.com/recovering-rare-earth-elements-from-phosphogypsum-as-a-secondary-resource-pathway/>...
6. [Rm11] Tunisian PG radiological characterization: Ra-226 214-970 Bq/kg; U-238 88.5; Pb-210 755.4; co-precipitation risk — https://www.researchgate.net/publication/24188010_Characterization_of_phosphogypsum_wastes_associated_with_phosphogypsum
7. [Rm12] IAEA / EU BSS: 1 Bq/g (1,000 Bq/kg) exemption threshold for U-238/Ra-226; TENORM classification — <https://www.mdpi.com/2071-1050/17/8/3473...>
8. [Rm17] GCT throughput (~6.5 Mt phosphate rock/yr; >10 Mt/yr PG at peak; 'hundreds of millions of tons' discharged at Gabès) — http://www.gct.com.tn/fileadmin/user_upload/Downloads/Rapport_Annuels/GCT-ANNUAL-REPORT-2012-PART1.pdf...
9. [Rw1] Largest SWRO plants: Ras Al-Khair 1,036,000 m³/d; Taweelah 909,200 (\$847M-1.2B); Sorek II 672,000; Rabigh 3 600,000 — <https://ide-tech.com/en/project/sorek-b-desalination-plant/>...
10. [Rw2] SWRO CAPEX \$800-1,200 per m³/day for mega-plants (>100k m³/d) — <https://supra-international.com/insights/seawater-reverse-osmosis-swro-desalination-technology-comprehensive-a...>
11. [Rw3] LCOW at plant gate: \$0.40-0.60/m³ (Jubail 3A \$0.41; Sorek II bid \$0.32-0.45); energy share 40-60% of OPEX — <https://pmc.ncbi.nlm.nih.gov/articles/PMC8953854/...>
12. [Rw4] Pumping physics: 400 m lift = 1.09 kWh/m³ at 100% eff; 1.45 at 75%; friction 0.5-1.5 kWh/m³ over 300 km; realistic 2.0-2.5 kWh/m³ — https://www.engineeringtoolbox.com/water-pumping-costs-d_1527.html...
13. [Rw5] Jubail-Buraydah IWTP: 587 km, 650,000 m³/d, \$2.26B (\$3.85M/km); twin-pipe scale \$5-7M/km — <https://www.meed.com/saudi-arabia-awards-22bn-water-transmission-contract...>
14. [Rw6] Delivered inland desalinated cost \$1.10-1.50/m³ (plant gate + transport + pumping energy) — <https://investriyadh.ai/intelligence/water-scarcity-investment/>...
15. [Rw7] Tunisian water tariffs: SONEDE 0.200-2.310 TND/m³; irrigation 0.060-0.160 TND/m³; groundwater 0.112-0.162 TND/m³ — <https://www.sonede.com.tn/tout-savoir-sur-leau/la-facture-deau-methodes-de-paiement/>...
16. [Rw13] Grid constraints: 3 GW overwhelms local 400 kV; new lines \$1-2.5M/km; renewables <5% of generation; curtailment risk — <https://reglobal.org/tunisia-focuses-on-grid-expansion-for-integrating-renewable-energy/>...
17. [Rc2] Gafsa basin: DInSAR-documented subsidence at Moulares; geothermal gradient 25-30°C/km → 23-29°C at 100-300 m depth — https://www.researchgate.net/publication/251701016_Phosphate_mine_subsidence_deduced_from_differential_interferometry
18. [Rc3] Gafsa radiation: 226Ra 16.3-375.1 Bq/kg in soils; phosphate tailings to 7,606 Bq/kg; radon-222 accumulation in enclosed workings; alpha-induced soft errors — https://www.researchgate.net/publication/380540277_Assessment_of_radiation_exposure_in_the_phosphate_mining_area
19. [Rc4] ASHRAE TC9.9 Class H1 (5th ed.): 18-22°C recommended, 25°C max for AI accelerators; AI datacenter CAPEX \$20-35M/MW (incl. IT), \$8-15M/MW facility-only — <https://envigilance.com/compliance/ashrae-tc-9-9/>...
20. [Rc5] GPU cost: ~100,000 B200-class per 100 MW; \$25-40k each → \$2.5-4B per 100 MW; 300 MW ≈ \$7.5-12B — <https://intuitionlabs.ai/articles/data-center-gpu-pricing-2026...>
21. [Rc6] US BIS export controls 2024-26: UAE 35,000 GPUs (G42/HUMAIN, 'Compute Diplomacy'); Saudi case-by-case 35,000 cap; Tunisia has no bilateral AI-security agreement — <https://www.commerce.gov/news/press-releases/2025/11/statement-uae-and-saudi-chip-exports...>
22. [Ru1] Uranium from wet-process phosphoric acid: Freeport (US) produced 14M lbs U₃O₈ 1978-99 at >95% DEHPA/TOPO recovery; ~20% of US uranium in the 1980s came from phosphates — <https://www.osti.gov/servlets/purl/1357599...>
23. [Ru3] WPA uranium SX cost \$59-83/lb U₃O₈ (IX \$46-76) vs \$80-95/lb 2024-26 spot — <https://repositories.lib.utexas.edu/bitstreams/89897275-85cf-4cd6-a0c5-1d8daa353799/download...>
24. [Ru4] REE from phosphoric acid: PhosAgro pilot; D2EHPA extracts >85% of heavy REEs (Dy, Y, Er) from WPA — <https://www.mdpi.com/2673-4117/7/2/58...>
25. [Ru5] Gafsa phosphate rock uranium content 45-140 ppm (southern basin) — https://www.researchgate.net/publication/380540277_Assessment_of_radiation_exposure_in_the_phosphate_mining_area